outer ones, just as Copernicus had deduced for the motion of the planets about the Sun. He found that Mercury and Venus went through phases like the Moon (showing they orbited the Sun). Moreover, the cratered Moon and the spotted Sun challenged the perfection of the heavens. This may in part constitute the sort of trouble Tertullian was worried about thirteen hundred years earlier, when he pleaded, “If you have any sense or modesty, have done with prying into the regions of the sky, into the destiny and secrets of the universe.”

In contrast, Galileo taught that we can interrogate Nature by observation and experiment. Then, “facts which at first sight seem improbable will, even on scant explanation, drop the cloak which had hidden them and stand forth in naked and simple beauty.” Are not these facts, available even for skeptics to confirm, a surer insight into God’s Universe than all the speculations of the theologians? But what if these facts contradict the beliefs of those who hold their religion incapable of making mistakes? The princes of the Church threatened the aged astronomer with torture if he persisted in teaching the abominable doctrine that the Earth moved. He was sentenced to a kind of house arrest for the remainder of his life.

A generation or two later, by the time Isaac Newton demonstrated that simple and elegant physics could quantitatively explain—and predict—all the observed lunar and planetary motions (provided you assumed the Sun at the center of the Solar System), the geocentrist conceit eroded further.

In 1725, in an attempt to discover stellar parallax, the painstaking English amateur astronomer James Bradley stumbled on the aberration of light. The term “aberration,” I suppose, conveys something of the unexpectedness of the discovery. When observed over the course of a year, stars were found to trace little ellipses against the sky. But all the stars were found to do so. This could not be stellar parallax, where we would expect a big parallax for nearby stars and an indetectible one for faraway stars. Instead, aberration is similar to how raindrops falling directly down on a speeding auto seem to the passengers to be falling at a slant; the faster the car goes, the steeper the slant. If the Earth were stationary at the center